

Veer Phatak

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PROFILE

Final-year MEng Aeronautics and Astronautics student specialising in Aerodynamics at the University of Southampton, graduating July 2027 with a 71.7% first-class Year 3 average. Team leader and aerodynamic design lead on a final-year project investigating the cornering aerodynamics of a MotoGP-style motorcycle through CFD and wind tunnel testing, also leading CAD development and model manufacture. Experienced across the aero development loop: ANSYS Fluent CFD, wind tunnel testing correlated against CFD and XFLR5, a first-class experimental fluid dynamics project with a bespoke computer-vision measurement system, and an R&D internship at TREND Networks where a redesigned part was approved for manufacture.

CORE SKILLS

Aerodynamics and CFD: ANSYS Fluent, XFLR5, XFOIL, OpenVSP/VSPAERO, panel methods (Python), aerofoil and boundary layer analysis, aerodynamic trade studies

Wind tunnel and experimental: wind tunnel testing, CFD and experiment correlation, test rig design and build, calibration, computer-vision measurement, uncertainty analysis

CAD and manufacture: SolidWorks, design for manufacture (injection moulding, additive manufacturing), rapid 3D-printed prototyping (Bambu Studio), fit and tolerance checks

FEA: ANSYS Mechanical, mesh convergence studies, stress and displacement interpretation

Programming and data: Python (NumPy, pandas, OpenCV, Matplotlib), MATLAB, Excel; data processing, visualisation and automation

AERODYNAMICS PROJECTS

Final-Year Group Design Project: Cornering Aerodynamics of a MotoGP-Style Motorcycle Sep 2026 - Present
Supervisors: Dr David Marshall and Dr Renan Soares

- **Team leader** of a student group investigating the cornering aerodynamics of a MotoGP-style motorcycle through CFD and wind tunnel testing, coordinating the team and liaising with supervisors.
- **Aerodynamic design lead**, responsible for aerodynamic design direction across the CFD and wind tunnel programme.
- **CAD development and manufacturing lead**, responsible for the CAD model and manufacture of the wind tunnel test model.

Individual Master's Project: Hydrodynamic Analysis of Large Plastics in Open-Channel Flow Oct 2025 - May 2026

Graded 75% (First). Supervisor: Dr Sergio Maldonado

- Designed and built a laboratory open-channel flume and paddlewheel drive (SolidWorks, Arduino stepper control) to reproduce turbulent flow, covering component selection, bearing and shaft mounting, sealing and camera illumination.
- Developed a Python/OpenCV measurement system (segmentation, contour fitting, nearest-neighbour tracking, Kalman filtering, homography calibration) reconstructing trajectories from 240 fps footage at **98.4% mean detection coverage and 1.5 mm positional accuracy**.
- Analysed **58 experimental runs** across spheres, rods and discs using velocity, slip, drag, dispersion and tortuosity, with Reynolds and Stokes number analysis comparing flume and water tunnel results.
- Diagnosed sealing, drive response and frame-rate issues, and validated every tracked run against raw footage to reject reflections and missed detections.

Fixed-Wing UAV Wing Design, Aerodynamics and Stability Lead (Group Project) 2024 - 2025

- **Ranked 1st in the cohort:** delivered the largest minimum-to-maximum speed range in the year group, **23.1 m/s predicted in XFLR5 against 19.8 m/s measured in flight**, using single-slot Fowler flaps and vortex generators.
- Compared planforms in XFLR5 and XFOIL within mass and volume limits, balancing lift, drag and stability under load.
- Led configuration selection using decision matrices, and managed Gantt and resource planning that delivered the project early.

EDUCATION

University of Southampton, MEng Aeronautics and Astronautics (Aerodynamics)

Sep 2023 - Jul 2027

- **Year 3: 71.7% average (First).** Aerospace Control Design 80%, Individual Project 75%, Aircraft Structural Design 75%, Advanced Aeronautics 74%, Aerothermodynamics 68%.
- **Wind tunnel laboratories:** measured NACA 0020 pressure distribution against angle of attack, boundary layer development and finite wing lift and drag, and correlated results against CFD, XFOIL and XFLR5, analysing discrepancies.
- **Advanced Aeronautics:** implemented a Python doublet panel method with accuracy and panel-distribution convergence studies.
- **FEA:** ANSYS Mechanical mesh convergence study from roughly 24,000 to 1.9 million elements, balancing peak stress accuracy against solve time and memory.

Reading School, Reading

Sep 2016 - Jul 2023

- A-levels: A* Economics, A Mathematics, A Physics, B Classical Civilisation. AS Additional Mathematics (A). 10 GCSEs including eight grade 9s.

ENGINEERING EXPERIENCE

TREND Networks, R&D Intern

Jul 2026 - Sep 2026

- **Handheld instrument enclosure:** designed from initial concept around a pre-manufactured PCB, fitting clearances, port positions and mounting points while reconciling requirements from several engineers. Refined over three iterations from a large, angular first version to a smaller, smoother design with protection for the test connections.
- **Battery compartment redesign:** redesigned an existing product's battery compartment for user-replaceable batteries under EU regulations within a fixed envelope, balancing battery protection, durability, styling and ease of use. Ten iterations produced a design **approved and released for manufacture**.
- **Ribbon fibre spool:** designed a spool to feed and protect fibre during testing, maximising stored fibre length within each tester case over three iterations.
- Ran all office 3D printing, prototyping each iteration the same day to check fit, wall thickness, warping and tolerances against the CAD, with injection moulding in mind for production parts.

Southampton University Formula Student (SUFST), Race Engineering and Chassis

2023 - Present

- Researched front roll hoop integration into the chassis and presented technical and feasibility findings to the department and team management.
- Built Python and MATLAB lap-time and tyre degradation simulations to evaluate setup trade-offs, contributing to **2nd in the FSUK Simulator Championship and 10th overall at FSUK 2025**.

LEADERSHIP AND WORK EXPERIENCE

Head Coach, Boyne Hill Cricket Club

May 2025 - Present

- Plan and lead weekly two-hour training sessions for U9 girls and U13 boys teams.

Other experience: Private Tutor, ages 8 to 14 (2022 - 2025); Front of House, Seasonality Restaurant (2022 - 2024); Dynamos Cricket Coach, ECB programme (2021 - 2022); Tutor, Brilliant Tutors Academy (2022); Sales Assistant and Till Manager, Kay's Chemist (2021 - 2022).

ACHIEVEMENTS AND INTERESTS

- 2nd overall, FSUK lap time challenge and Simulator Championship 2025.
- Boyne Hill CC 1st XI wicketkeeper-batter with 16 years of competitive cricket; captained Southampton University CC indoor team and outdoor 3rd XI.
- ECB and Berkshire qualified cricket coach; qualified first aider; Sports Prefect at Reading School.

References available on request.